

# SPHERE: MISSION CONTROL

## INSTRUCTOR MANUAL, SAFETY GUIDANCE AND ASSESSMENT KEY

**Scope:** The activity is intended for introductory undergraduate engineering or physics teaching. Instructors should adapt the risk assessment, supervision ratio and assessment weighting to local institutional requirements.

### 1. LEARNING OUTCOMES

On completion, students should be able to:

- Distinguish conduction, convection and radiation in the test system.
- Apply Newton's law of cooling,

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt},$$

and interpret the fitted cooling constant  $k$  ( $\text{min}^{-1}$ ).

- Acquire and plot temperature data ( $T_{\text{obs}}$ ) at defined time intervals ( $t = 0$  to  $15 \text{ min}$ ).
- Calculate residuals ( $|\Delta T| = |T_{\text{obs}} - T_{\text{model}}|$ ) and MAE ( $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\Delta T_i|$ ), then discuss uncertainty and limitations of the lumped-parameter model.



## 2. 60-MINUTE LESSON ROADMAP

| TIMELINE  | LESSON PHASE                | WHAT YOU & YOUR STUDENTS ARE DOING                                                                                                                                                                                                                |
|-----------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0–10 min  | 1. Theory and concept check | Review the three heat-transfer modes, the water-bath boundary condition ( $T_{\text{env}} = 0.0^{\circ}\text{C}$ ) and the limits of the spaceflight analogy. Complete the preliminary concept check.                                             |
| 10–20 min | 2. Capsule Assembly         | Assign teams their insulation setup (A: bare, B: cotton, C: reflective film, D: multilayer assembly). Demonstrate probe placement and the pre-immersion leak test.                                                                                |
| 20–35 min | 3. Fifteen-minute test      | Add 100 mL of water at $80.0^{\circ}\text{C}$ to each capsule. Measure the bath temperature, immerse the capsules, start the timer and record temperatures every 60 seconds.                                                                      |
| 35–48 min | 4. Analysis                 | Plot measured ( $T_{\text{obs}}$ ) and predicted ( $T_{\text{model}}$ ) temperatures, calculate residuals and MAE ( $\text{MAE} = \frac{1}{n} \sum  \Delta T_i $ ), and compare the error with instrument accuracy ( $\pm 0.5^{\circ}\text{C}$ ). |
| 48–60 min | 5. Technical discussion     | Compare the four configurations and discuss material mechanisms, model assumptions, uncertainty and design trade-offs.                                                                                                                            |



## 3. LAB SETUP & SAFETY CHECKLIST

Complete and document the following controls before the session:

- **Hot-water control:** Do not use boiling water. Prepare water at  $80.0^{\circ}\text{C}$  and use heat-resistant gloves, eye protection and a stable pouring area in accordance with the local risk assessment.
- **Bath condition:** Prepare an ice-water slurry and measure its temperature. Do not assume an exact  $0.0^{\circ}\text{C}$  boundary without verifying.
- **Leak Prevention:** Remind students to hand-tighten the PG7 gland dome nut so the internal rubber ring clamps securely around the wire.
- **Probe Placement:** Check that each probe hangs in the center of the liquid and isn't pressed against the plastic jar walls.

## 4. REFERENCE-MODEL OUTPUT

The table lists outputs from the assumed coefficients at  $t = 15 \text{ min}$ , with  $T_0 = 80.0^\circ\text{C}$  and  $T_{\text{env}} = 0.0^\circ\text{C}$ . These are model values ( $T_{\text{model}}$ ), not acceptance criteria for measured results.

| INSULATION TYPE        | ASSUMED COOLING CONSTANT ( $k$ , $\text{min}^{-1}$ ) | MODEL TEMP AT 15 MIN                     | QUALITATIVE INTERPRETATION                                                                                                        |
|------------------------|------------------------------------------------------|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| A. Bare Control        | $k \approx 0.150 \text{ min}^{-1}$                   | $8.0^\circ\text{C} - 12.0^\circ\text{C}$ | Rapid heat loss through unshielded conduction into water and internal liquid circulation.                                         |
| B. Cotton Sleeve       | $k \approx 0.080 \text{ min}^{-1}$                   | $\approx 24.1^\circ\text{C}$             | Cotton fibers trap pockets of still air, which is a poor heat conductor.                                                          |
| C. Mylar Foil          | $k \approx 0.040 \text{ min}^{-1}$                   | $\approx 43.9^\circ\text{C}$             | Shiny metallic surface reflects infrared heat radiation back into the core.                                                       |
| D. Multilayer assembly | $k \approx 0.015 \text{ min}^{-1}$                   | $\approx 63.9^\circ\text{C}$             | The combined layers increase effective thermal resistance through trapped air, reduced fluid motion and lower radiative exchange. |

### SECTION 5: OFFICIAL SOLUTIONS & TEACHER ANSWER KEY

#### ANSWER KEY

#### Question 1: Why doesn't Mylar foil work by itself in cold water?

**Sample answer:** A reflective film can reduce radiative exchange, but it has little thickness and therefore provides limited resistance to conduction when it is in direct contact with the capsule and surrounding water. A low-conductivity spacer introduces trapped air and additional thermal resistance.

*Marking guidance: award credit for distinguishing radiative performance from conductive resistance.*

#### Question 2: What does the cooling constant ( $k$ ) tell us?

**Sample answer:** Within the same test geometry and boundary conditions, a smaller fitted  $k$  means a slower exponential approach to  $T_{\text{env}}$  ( $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$ ). The value is an effective system parameter with units of  $\text{min}^{-1}$ ; it is not the material thermal conductivity  $\lambda$ .

### Question 3: Why can the multilayer assembly reduce cooling?

**Sample answer:** The materials introduce complementary resistance mechanisms:

- **Cotton** traps air and reduces effective conduction.
- **Bubble wrap** limits air movement within sealed cells.
- **Reflective film** reduces radiative exchange when an air gap is present.

Interfaces between layers also add contact resistance. The measured result must be used to determine the actual performance ( $k \approx 0.015 \text{ min}^{-1}$ ); no specific final temperature should be assumed without empirical data.

### Question 4: Where could experimental errors come from?

**Sample Answer (Accept any two sensible answers):**

1. The sensor tip was touching the cold plastic jar wall instead of hanging in the middle of the water.
2. Water leaked into the jar through a loose cable gland.
3. The ice bath warmed up above  $0.0^\circ\text{C}$  during the 15 minutes.
4. The water inside the jar wasn't stirred, so warm water rose to the top and cold water sank to the bottom.

### Question 5: What happens if your ice bath warms up?

**Sample Answer:** Students should measure the real bath temperature (e.g.  $T_{\text{env}} = 4.5^\circ\text{C}$ ) and type that new number into the  $T_{\text{env}}$  box in the Mission Control simulator. Updating the environment temperature makes the theoretical curve ( $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$ ) match the real classroom conditions.

*Grading Tip: Award credit if they mention updating  $T_{\text{env}}$  or the environment temperature input in the software.*

### Question 6: What if we double the water volume?

**Sample Answer:** Doubling the water from 100 mL to 200 mL doubles the mass of hot liquid ( $Q = m \cdot c \cdot \Delta T$ ), meaning it holds twice as much thermal energy. Because there is more heat to lose through roughly the same surface area, it cools down **slower**, and the cooling constant  $k$  **decreases**.

*Grading Tip: Look for: cools slower and  $k$  decreases.*

### Question 7: Spacesuit Design Trade-Offs

**Sample Answer:** You can't just add 50 layers because the spacesuit would become too heavy, bulky, and stiff. Astronauts wouldn't be able to bend their elbows or fingers during a 7-hour spacewalk. Also, human bodies generate lots of body heat when working hard - too much insulation would trap that heat and cause dangerous overheating without heavy water-cooling systems.

*Grading Tip: Accept answers discussing joint flexibility/mobility, suit weight/bulk, or body heat buildup.*

### Question 8: How can the MAE be reduced?

**Sample Answer:**

1. Tighten the cable gland securely to prevent any cold water leaks.
2. Position the probe consistently near the centre of the liquid without contacting the wall ( $\pm 0.5^{\circ}\text{C}$ ).
3. Swirl the jar gently before reading to keep the water mixed evenly.
4. Take readings at consistent recorded times ( $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |T_{\text{obs},i} - T_{\text{model},i}|$ ).

## 6. GRADING RUBRIC & CHECKLIST

| CRITERIA           | EXCELLENT (90-100%)                                                                              | GOOD (75-89%)                                                       | NEEDS WORK (<75%)                                       |
|--------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------|
| Data Logging       | All 15 minutes filled with accurate readings, absolute differences, and helpful notes.           | Completed with only 1-2 missed minutes; minor calculation mistakes. | Multiple blank rows; missing calculations.              |
| Graphing           | Clean, accurate points plotted with a smooth line; dashed theoretical curve clearly shown.       | Points plotted with minor curve scaling inaccuracies.               | Points not connected; axes mislabeled or missing lines. |
| Analysis Questions | Clear, thoughtful answers explaining conduction, convection, radiation, and MLI synergy (Q1-Q8). | Good understanding of heat transfer with minor details missing.     | Confuses conduction with radiation; incomplete answers. |